```
classify_button.on_click(on_classify_click)
display(widgets.VBox([title, label, text_area, classify_button, output]))
```

```
VBox(children=(HTML(value="<h1 style = 'font-weight:800'>Researchly Research Abstract Classifier</h1><p>This a…
```

Question 3

3.1

In [21]:
```
import os

CORPUS_DIR = "research_corpus"

records = []
for fname in sorted(os.listdir(CORPUS_DIR)):
    if fname.endswith(".txt"):
        with open(os.path.join(CORPUS_DIR, fname), "r", encoding = "utf-8") as f:
            content = f.read()
            records.append({"filename": fname, "content": content})

articles_data = pd.DataFrame(records).sort_values("filename").reset_index(drop = True)
print("Shape:", articles_data.shape)
articles_data[["filename"]].assign(characters = articles_data["content"].str.len())
```

```
Shape: (5, 2)
```

Out[21]:

	filename	characters
0	research_1.txt	38160
1	research_2.txt	16104
2	research_3.txt	26154
3	research_4.txt	38950
4	research_5.txt	26831

In [22]:
```
articles_data.head()
```

Out[22]:

	filename	content
0	research_1.txt	The purposes of this article are to position m...
1	research_2.txt	Now I see it, now I don't:\nresearcher's posit...
2	research_3.txt	DEVELOPMENTAL RESEARCH: STUDIES OF\nINSTRUCTIO...
3	research_4.txt	Abstract\nCronbach's alpha is a statistic comm...
4	research_5.txt	Denzin's Paradigm Shift: Revisiting Triangulat...

3.2

In [23]:
```
for _, row in articles_data.iterrows():
    text = row["content"]
    probe = text[:300]
    second_occurrence = text.find(probe, 300)
    flag = "DUPLICATED" if second_occurrence != -1 else "ok"
    print(f"{row['filename']:16s} len = {len(text):6d} {flag}")
```

```
research_1.txt   len =  38160 DUPLICATED
research_2.txt   len =  16104 ok
research_3.txt   len =  26154 ok
research_4.txt   len =  38950 ok
research_5.txt   len =  26831 ok
```

In [24]:
```
def remove_duplicated_body(text):
    """Detect an exact first-half/second half repeat and keep only one copy"""
    n = len(text)
    half = n // 2
    first_half, second_half = text[:half], text[half:]
    if len(first_half)> 500 and first_half.strip() == second_half.strip()[:len(first_half.strip())]:
        return first_half
    return text

def denoise_for_sentences(text):
    t = remove_duplicated_body(text)
    t = t.replace("\r", " ")
    t = re.sub(r"-\n", "", t)
```

```
    t = re.sub(r"\n+", " ", t)
    t = re.sub(r"\s+", " ", t).strip()
    return t

def denoise_text_articles(text):
    t = denoise_for_sentences(text)
    t = t.lower()
    t = re.sub(r"http\S+|www\.\S+", " ", t)
    t = re.sub(r"\([^)]*\d{4}[^)]*\)", " ", t)
    t = re.sub(r"\b[a-z]+\s*&\s*[a-z]+,?\s*\d{4}\b", " ", t)
    t = re.sub(r"\bp{1,2}\.\s*\d+\b", " ", t)
    t = re.sub("[^a-z\s]", " ", t)
    t = re.sub(r"\s+", " ", t).strip()
    tokens = word_tokenize(t)
    tokens = [lemmatizer.lemmatize(tok)
              for tok in tokens
              if tok not in stop_words and len(tok) > 2]
    return " ".join(tokens)

articles_data["Clean_Sentences"] = articles_data["content"].apply(denoise_for_sentences)
articles_data["Clean_Content"] = articles_data["content"].apply(denoise_text_articles)

print("Characters before vs After:")
for _, row in articles_data.iterrows():
    print(f"{row['filename']:16s} raw = {len(row['content']):6d} clean_sentences = {len(row['Clean_Sentences']):6d}")
```

```
Characters before vs After:
research_1.txt   raw =  38160 clean_sentences =  19079
research_2.txt   raw =  16104 clean_sentences =  16104
research_3.txt   raw =  26154 clean_sentences =  26154
research_4.txt   raw =  38950 clean_sentences =  38897
research_5.txt   raw =  26831 clean_sentences =  26830
```

3.3

In [25]:

```
from nltk.tokenize import sent_tokenize
from sklearn.feature_extraction.text import TfidfVectorizer

def freq_based_summarize(clean_sentence_text, n_sentences = 5):
    sentences = sent_tokenize(clean_sentence_text)
```

```
    sentences = [s.strip() for s in sentences
                 if len(s.split()) >= 6]
    n_sentences = min(n_sentences, len(sentences))

    vec = TfidfVectorizer(stop_words = "english")
    tfidf_matrix = vec.fit_transform(sentences)
    nnz_per_row = np.diff(tfidf_matrix.indptr)
    nnz_per_row[nnz_per_row == 0] = 1
    sentence_scores = np.asarray(tfidf_matrix.sum(axis = 1)).ravel() / nnz_per_row

    ranked = sorted(((sentence_scores[i], i, s)
                     for i, s in enumerate(sentences)), reverse = True)
    top = sorted(ranked[:n_sentences], key = lambda x: x[1])
    summary = " ".join(s for _, _, s in top)
    return summary, sentences, sentence_scores, ranked

first_row = articles_data.iloc[0]
summary, sentences, scores, ranked = freq_based_summarize(first_row["Clean_Sentences"], n_sentences = 5)
print("Article:", first_row["filename"], " | total candicate sentences:", len(sentences))
print()
print("Frequency-based (TF-IDF) summary (top 5 sentences): \n")
print(summary)
```

```
Article: research_1.txt  | total candicate sentences: 76

Frequency-based (TF-IDF) summary (top 5 sentences):

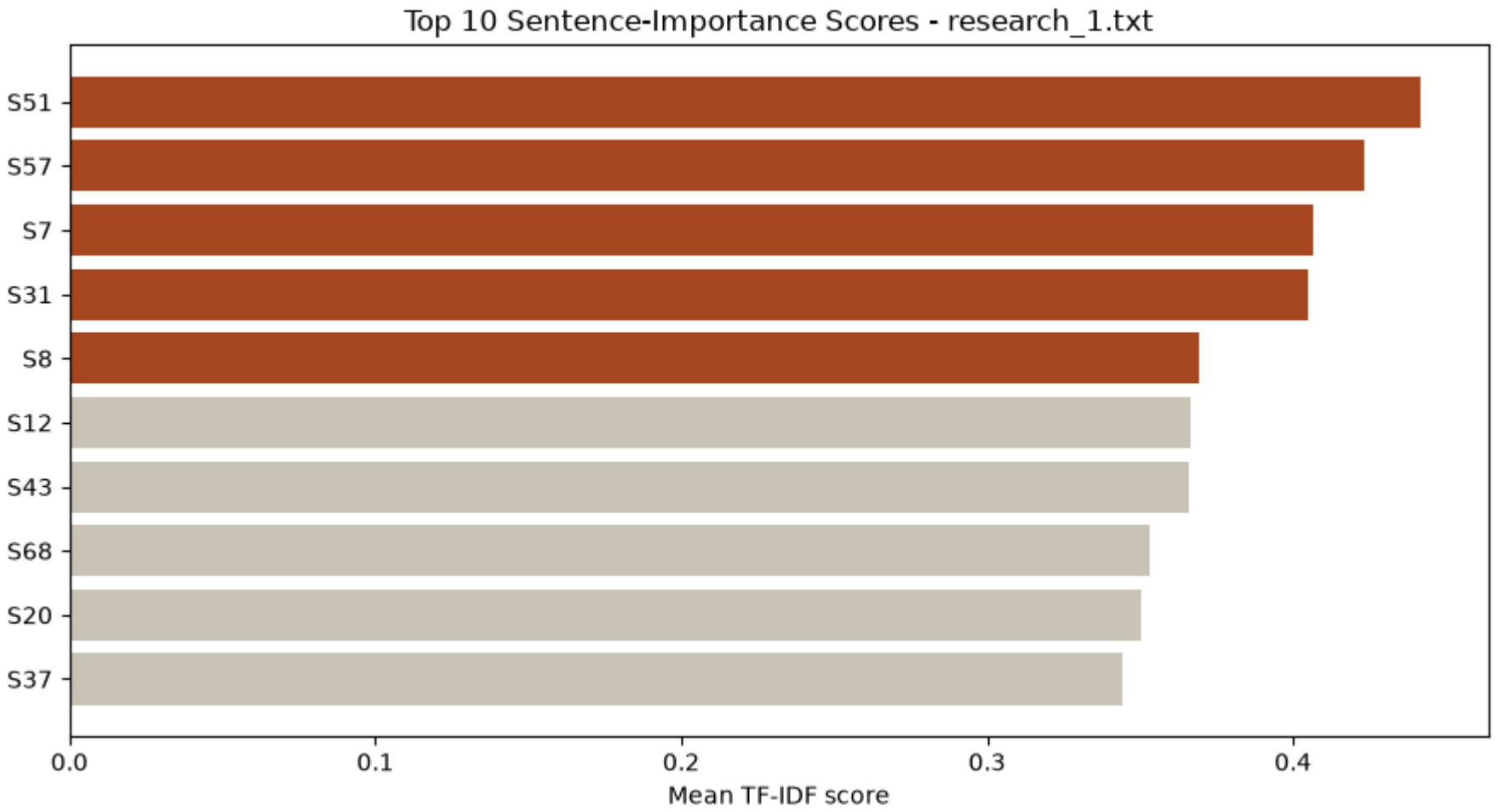
Further, they contend that the observer is separate from the entities that are subject to observation. Quantitative purists mai
ntain that social science inquiry should be objective. We will try to clarify the most important issues in the remainder of thi
s article. There are several interesting myths that appear to be held by some purists. Qualitative researchers also are not imm
une from constructive criticism.
```

In [26]:

```
top10 = sorted(((scores[i], i, s) for i, s in enumerate(sentences)), reverse = True)[:10]
selected_idx = {idx for _, idx, _ in sorted(ranked[:5], key = lambda x: x[1])}
labels = [f"S{idx + 1}" for _, idx, _ in top10]
values = [sc for sc, _, _ in top10]
colors = ["#a8481f" if idx in selected_idx else "#c9c3b6" for _, idx, _ in top10]

fig, ax = plt.subplots(figsize = (9, 5))
ax.barh(labels[::-1], values[::-1], color = colors[::-1])
ax.set_xlabel("Mean TF-IDF score")
```

```
ax.set_title(f"Top 10 Sentence-Importance Scores - {first_row['filename']}")
plt.tight_layout()
plt.show()
```



3.4

In [27]:
```
import networkx as nx
from sklearn.metrics.pairwise import cosine_similarity

remaining_data = articles_data.iloc[1:].reset_index(drop = True)
```

```
def textrank_aggregate_summarize(data, text_col = "Clean_Sentences", n_sentences = 8, min_words = 6):
    pool = []
    for _, row in data.iterrows():
        for s in sent_tokenize(row[text_col]):
            s = s.strip()
            if len(s.split()) >= min_words:
                pool.append((row["filename"], s))

    sentences = [s for _, s in pool]
    sources = [f for f, _ in pool]
    n_sentences = min(n_sentences, len(sentences))

    vec = TfidfVectorizer(stop_words = "english")
    sent_vectors = vec.fit_transform(sentences)
    sim_matrix = cosine_similarity(sent_vectors)
    np.fill_diagonal(sim_matrix, 0)

    graph = nx.from_numpy_array(sim_matrix)
    scores = nx.pagerank(graph)

    ranked = sorted(((scores[i], i, s) for i, s in enumerate(sentences)), reverse = True)
    top = sorted(ranked[:n_sentences], key = lambda x: x[1])
    summary = " ".join(s for _, _, s in top)
    return summary, sentences, sources, scores, ranked

summary_34, sentences_34, sources_34, score_34, ranked_34 = textrank_aggregate_summarize(remaining_data, n_sentences = 8)
print("Articles summarized as an aggregate:", remaining_data["filename"].tolist())
print("Total candidate sentences pooled:", len(sentences_34))
print()
print("Text Rank aggregate summary (Top 8 sentences, research 2, research 3, research 4, research 5):\n")
print(summary_34)
```

```
Articles summarized as an aggregate: ['research_2.txt', 'research_3.txt', 'research_4.txt', 'research_5.txt']
Total candidate sentences pooled: 559

Text Rank aggregate summary (Top 8 sentences, research 2, research 3, research 4, research 5):

In its simplest form, developmental research can be either  the study of the process and impact of specific instructional desi
gn and development efforts; or  a situation in which someone is performing instructional design, development, or evaluation ac
tivities and studying the process at the same time; or  the study of the instructional design, development, and evaluation pro
cess as a whole or of particular process components. The purposes of this chapter are to  explore the nature and background of
developmental research;  describe the major types of developmental research by examining a range of representative projects; 
analyze the methodological approaches used in the various types of developmental research;  describe the issues, findings, and
trends in recent developmental research; and  discuss the future of this type of research in our field. A Summary of the Two T
ypes of Developmental Research Type 1 Type 2 Emphasis Study of specific product or program design, development, &/or evaluation
projects Study of design, development, or evaluation processes, tools, or models Product Lessons learned from developing specif
ic products and analyzing the conditions that facilitate their use New design, development, and evaluation procedures &/or mode
ls, and conditions that facilitate their use Context-specific Generalized Conclusions     Conclusions from exploratory studies
through (formative and summative) evaluation studies" (p. 6). The Use of Cronbach's Alpha in Science Education Studies It is co
mmon to see the reliability of instruments used in published science education studies framed in terms of a statistic known as
Cronbach's alpha (Cronbach, 1951). Keywords: triangulation, Denzin, validity, reliability, confirmability, transferability, qua
litative research, multiple data source, multiple data collection methods, mitigating bias, data saturation, social change Intr
oduction Researchers often use a qualitative methodology to address social change despite the fact that most research does not
study social change, per se. Qualitative Research and Triangulation The application of triangulation (multiple sources of data)
can enhance the reliability of the study results (Stavros & Westberg, 2009) and enable one to saturate the data (P. Fusch & Nes
s, 2015). Denzin's Paradigm Shift: Triangulation in Qualitative Research Triangulation is an important concept regarding data a
nalysis for an empirical study. in a qualitative case study or ethnography would be within-method triangulation, whereas triang
ulating the data from a Fusch, Fusch, & Ness, 2018 Journal of Social Change 23 combination of quantitative and qualitative tech
niques in a mixed-methods study would be betweenmethod (or across method) triangulation.
```

In [28]:
```python
selected_idx_34 = sorted(ranked_34[:8], key = lambda x: x[1])
for score, idx, s in selected_idx_34:
    print(f"[{sources_34[idx]}] (score = {score:.4f})")
    print(" ", s)
    print()
```

[research_3.txt] (score = 0.0043)
In its simplest form, developmental research can be either the study of the process and impact of specific instructional de sign and development efforts; or a situation in which someone is performing instructional design, development, or evaluation activities and studying the process at the same time; or the study of the instructional design, development, and evaluation p rocess as a whole or of particular process components.

[research_3.txt] (score = 0.0034)
The purposes of this chapter are to explore the nature and background of developmental research; describe the major types of developmental research by examining a range of representative projects; analyze the methodological approaches used in the various types of developmental research; describe the issues, findings, and trends in recent developmental research; and di scuss the future of this type of research in our field.

[research_3.txt] (score = 0.0034)
A Summary of the Two Types of Developmental Research Type 1 Type 2 Emphasis Study of specific product or program design, deve lopment, &/or evaluation projects Study of design, development, or evaluation processes, tools, or models Product Lessons learn ed from developing specific products and analyzing the conditions that facilitate their use New design, development, and evalua tion procedures &/or models, and conditions that facilitate their use Context-specific Generalized Conclusions ⇒⇒⇒⇒ Conclusio ns from exploratory studies through (formative and summative) evaluation studies” (p. 6).

[research_4.txt] (score = 0.0036)
The Use of Cronbach’s Alpha in Science Education Studies It is common to see the reliability of instruments used in published science education studies framed in terms of a statistic known as Cronbach’s alpha (Cronbach, 1951).

[research_5.txt] (score = 0.0042)
Keywords: triangulation, Denzin, validity, reliability, confirmability, transferability, qualitative research, multiple data source, multiple data collection methods, mitigating bias, data saturation, social change Introduction Researchers often use a qualitative methodology to address social change despite the fact that most research does not study social change, per se.

[research_5.txt] (score = 0.0039)
Qualitative Research and Triangulation The application of triangulation (multiple sources of data) can enhance the reliabilit y of the study results (Stavros & Westberg, 2009) and enable one to saturate the data (P. Fusch & Ness, 2015).

[research_5.txt] (score = 0.0039)
Denzin’s Paradigm Shift: Triangulation in Qualitative Research Triangulation is an important concept regarding data analysis for an empirical study.

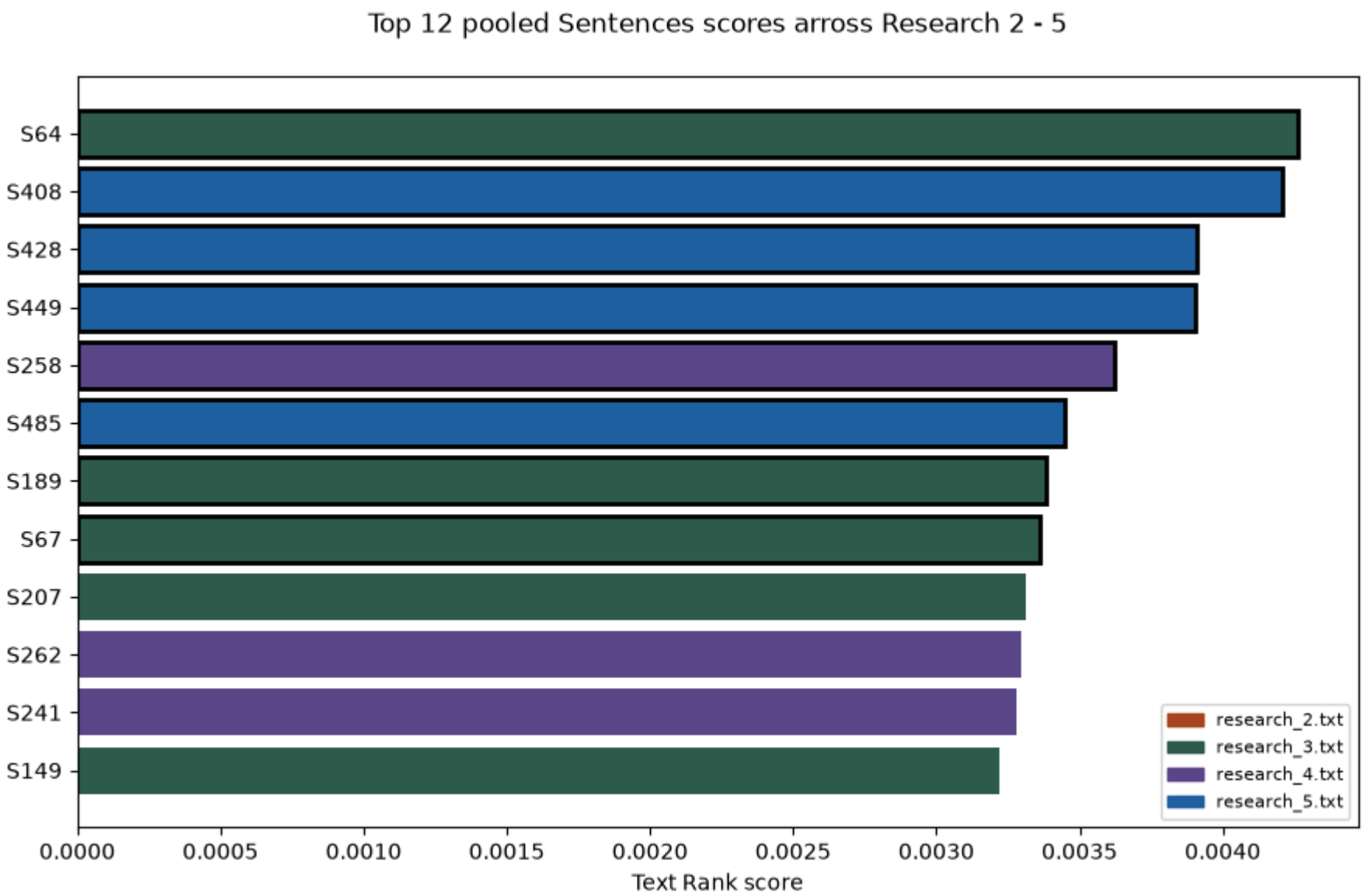
[research_5.txt] (score = 0.0034)
in a qualitative case study or ethnography would be within-method triangulation, whereas triangulating the data from a Fusch, Fusch, & Ness, 2018 Journal of Social Change 23 combination of quantitative and qualitative techniques in a mixed-methods study would be betweenmethod (or across method) triangulation.

In [29]:
```
top12_34 = ranked_34[:12]
selected_idx_set_34 = {idx for _, idx, _ in selected_idx_34}
article_order_34 = remaining_data["filename"].tolist()
palette_34 = {f: c for f, c in zip(article_order_34, ["#a8481f", "#2f5d4f", "#5b4a8a", "#2060a0"])}

labels_34 = [f"S{idx}" for _, idx, _ in top12_34]
values_34 = [sc for sc, _, _, in top12_34]
bar_colors_34 = [palette_34[sources_34[idx]] for _, idx, _ in top12_34]
edge_colors_34 = ["black" if idx in selected_idx_set_34 else "none" for _, idx, _ in top12_34]

fig, ax = plt.subplots(figsize = (9, 6))
ax.barh(labels_34[::-1], values_34[::-1], color = bar_colors_34[::-1], edgecolor = edge_colors_34[::-1], linewidth = 2)
ax.set_xlabel("Text Rank score")
ax.set_title("Top 12 pooled Sentences scores arross Research 2 - 5\n")

handles = [plt.Rectangle((0, 0), 1, 1, color = c) for c in palette_34.values()]
ax.legend(handles, palette_34.keys(), loc = "lower right", fontsize = 8)
plt.tight_layout()
plt.show()
```


Top 12 pooled Sentences scores arross Research 2 - 5
S64
S408
S428
S449
S258
S485
S189
S67
S207
S262
S241
S149
research_2.txt
research_3.txt
research_4.txt
research_5.txt
0.0000
0.0005
0.0010
0.0015
0.0020
0.0025
0.0030
0.0035
0.0040
Text Rank score

Question 4